
Scalability & Future Plan: Emotion Detection and Learning Support Engine

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Team ID:

Current System Limitations

S.No	Limitation	Impact	Priority
1	Supports only English text input	Limits accessibility for multilingual users	High
2	Limited emotion categories	Some complex emotions may not be identified accurately	High
3	CSV-based log storage	Suitable for small projects but not ideal for large-scale	Medium
4	Limited support for concurrent users	Performance may decrease with many simultaneous users	Medium
5	Web application only (no mobile app)	Users cannot access a native mobile experience	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Individual/Small-scale	Deploy on cloud infrastructure with load balancing
2	Data Storage	CSV files	Upgrade to MySQL, PostgreSQL, or MongoDB
3	Performance	Local Streamlit execution	Optimize model inference and deploy on cloud with GPU
4	Security	Basic application security	Add user authentication, encryption, and RBAC

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
	Cloud Deployment		Improve accessibility

Phase 1		Short Term (1–2 Months)	
Phase 2	Database & Authentication	Medium Term (2–4 Months)	Secure user management
Phase 3	Multilingual & Voice Support	Long Term (4–6 Months)	Increase usability
Phase 4	Advanced AI & Mobile App	Future (6+ Months)	Improve accuracy & experience

The current implementation successfully achieves the project objectives and provides an effective AI-powered emotion detection and learning support system. The proposed scalability plan focuses on improving performance, storage capacity, security, and deployment readiness. Future enhancements aim to transform the application into a robust, cloud-enabled, intelligent educational platform capable of supporting a larger user base while delivering more accurate and personalized learning assistance.